

HTML Transformation Workbench

Architecture, User Stories & Implementation Roadmap

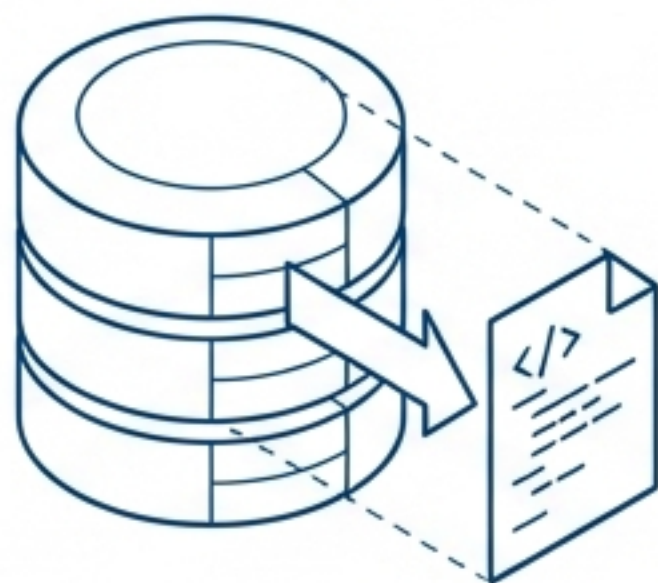
PROJECT: HTW-2024

SPEC VERSION: 1.0

STATUS: APPROVED

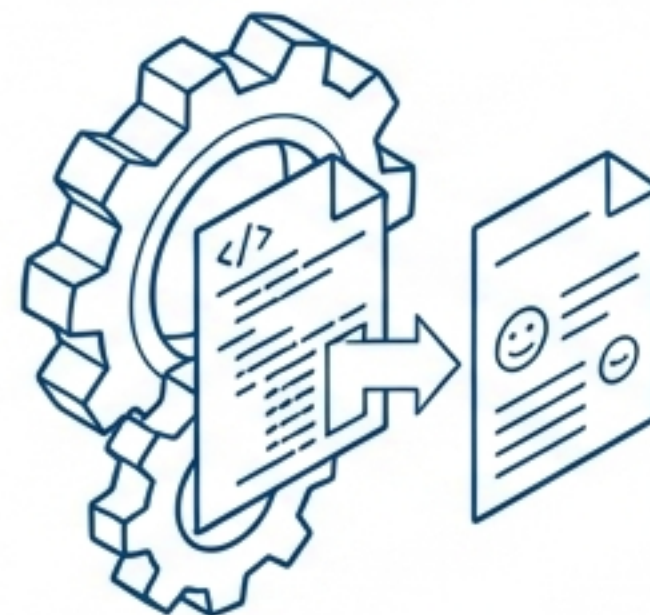
System Definition

A modular, plugin-based application for the complete lifecycle of cached HTML management. The system provides a unified visual interface for interacting with the **HTML Graph API**, replacing ad-hoc manual processes.



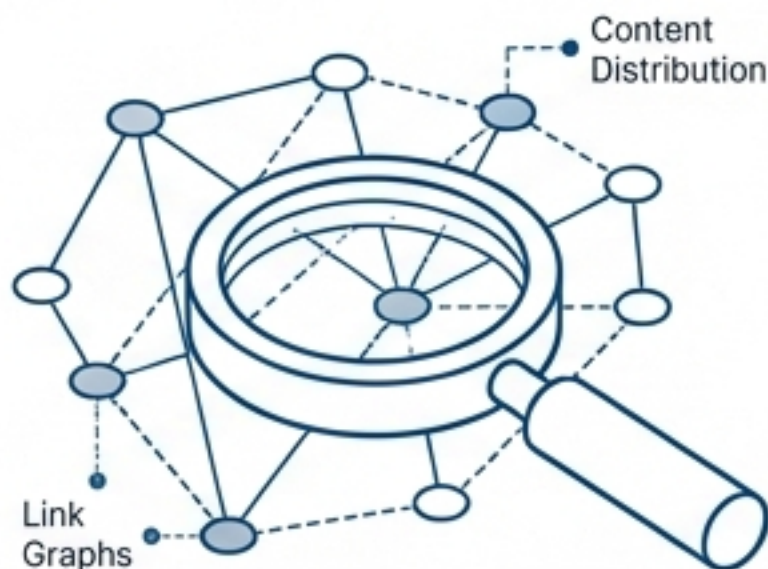
LOAD

Retrieve cached HTML content via unique cache keys.



TRANSFORM

Apply visual transformations: sentiment filtering, content extraction, and style simplification.



ANALYSE

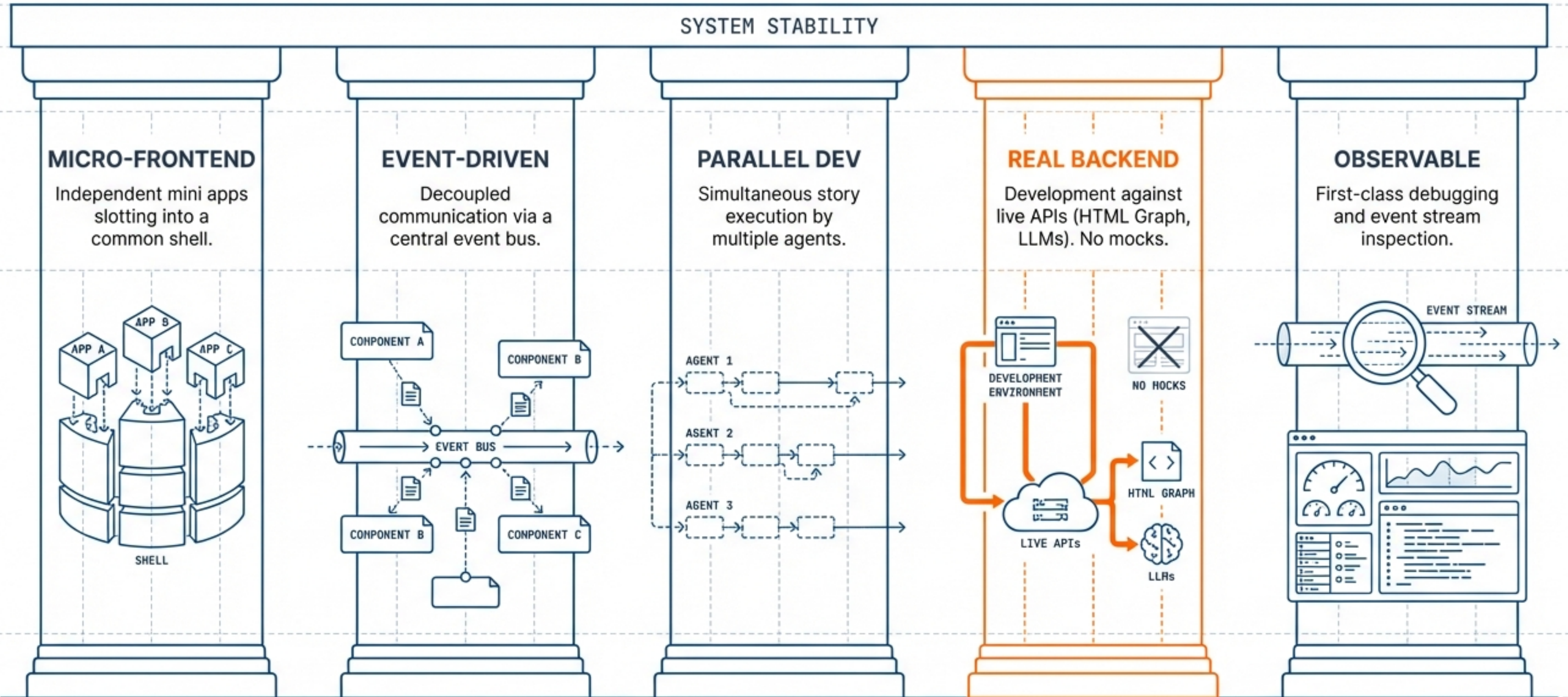
Inspect page structure, content distribution, and generate link graphs.



MANAGE

Handle transformation profiles and cached entities.

Key Design Principles



THE LAYERED ARCHITECTURE

LAYER 4: SHARED UI COMPONENTS

LAYER 2: MINI APPS

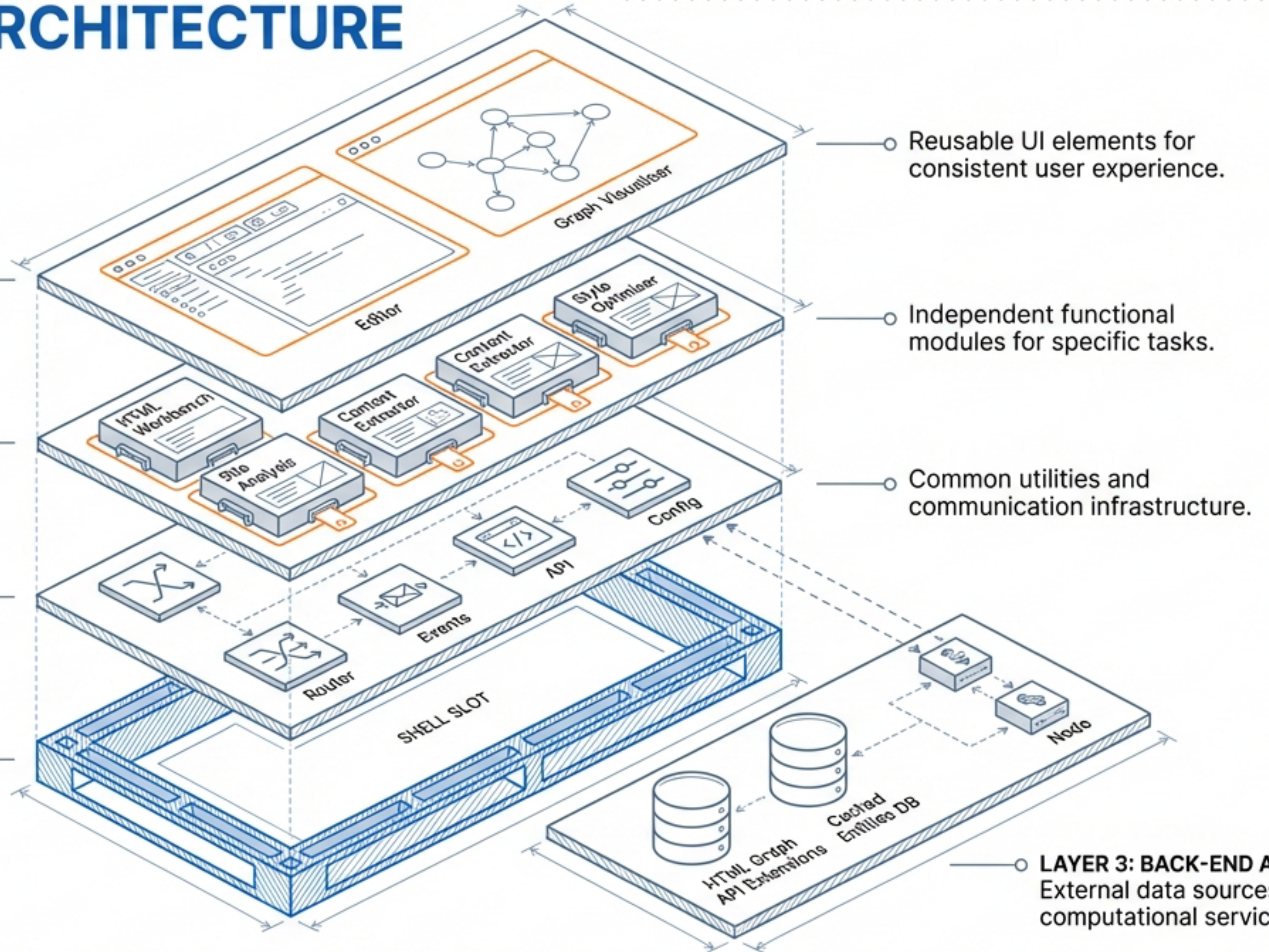
Pluggable module

LAYER 1: SHARED SERVICES

communication infrastructure.

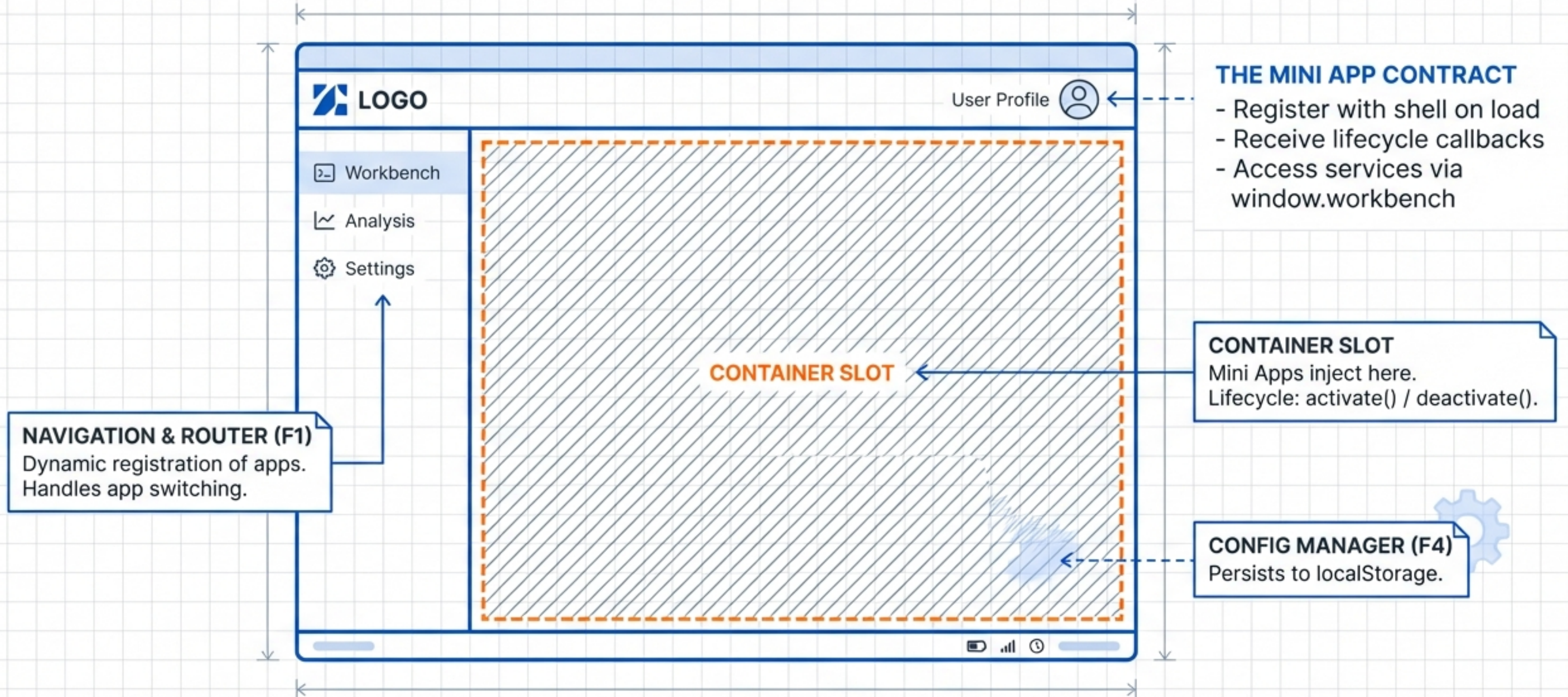
LAYER 0: FOUNDATION SHELL

Core application framework and container.



LAYER 0: THE FOUNDATION SHELL

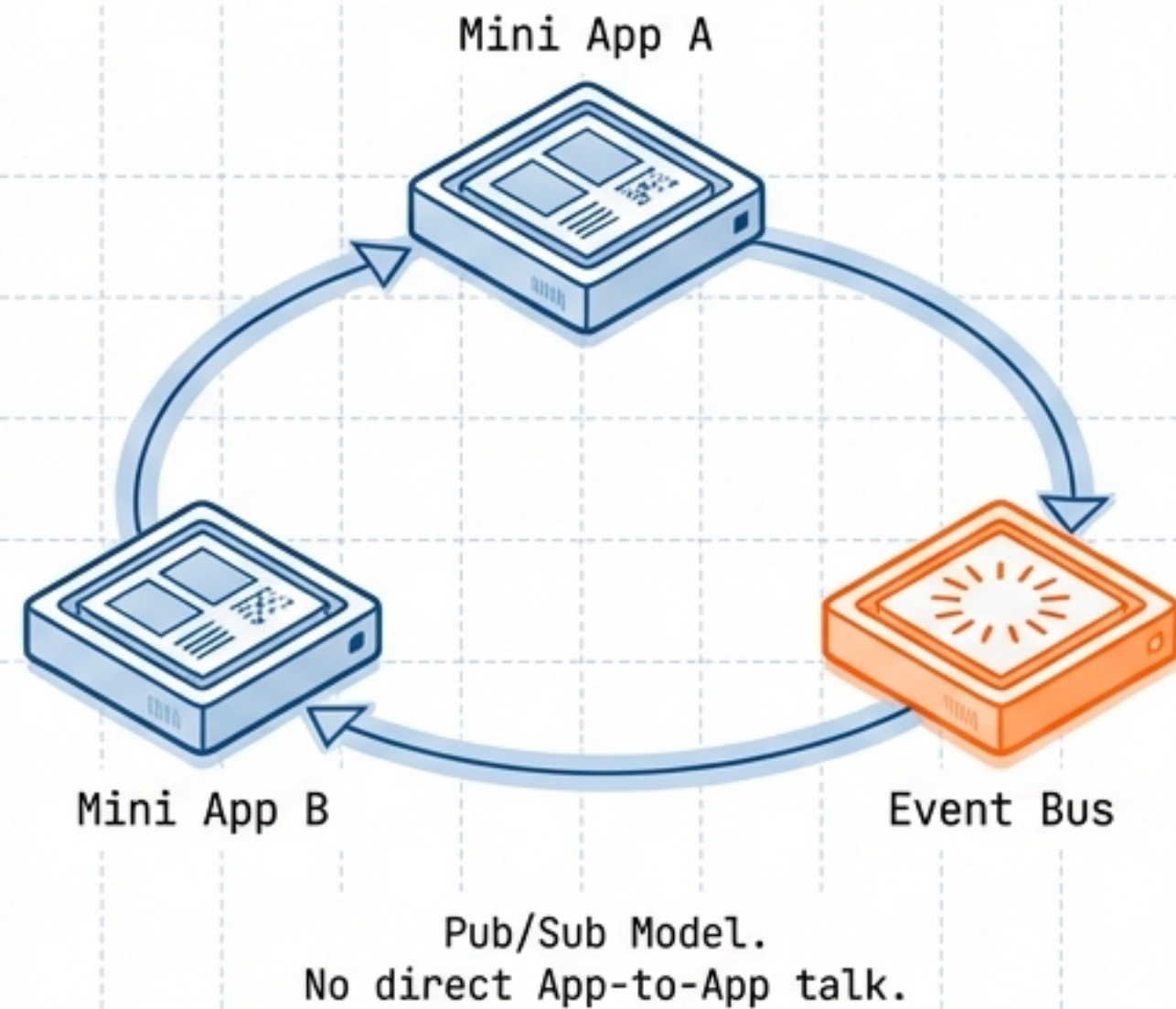
Stories F1, F4



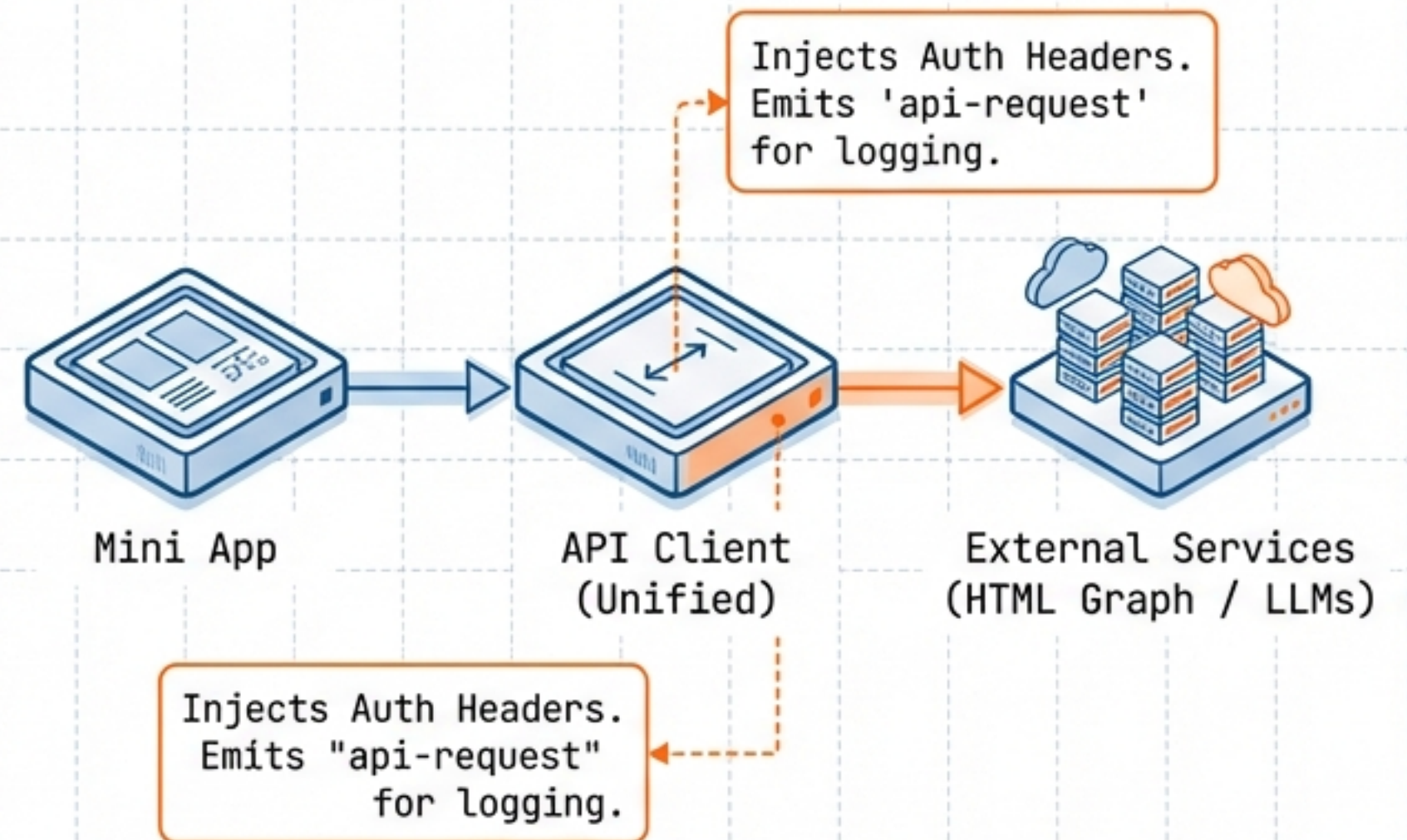
THE NERVOUS SYSTEM

Event Bus & APIs (Stories F2, F3)

INTERNAL: Event-Driven Architecture (F2)

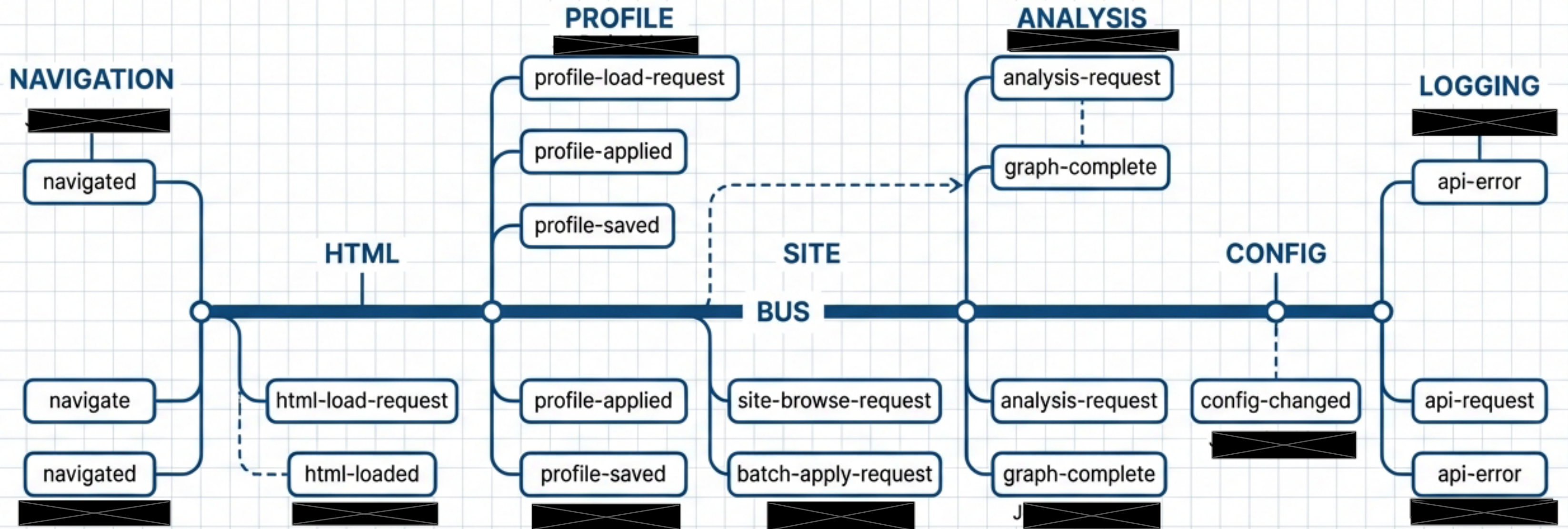


EXTERNAL: API Client Service (F3)



THE EVENT CATALOG

Internal System Language



BUILT-IN OBSERVABILITY & CONFIGURATION

Stories D1, D2, M1, M2

CONFIGURATION (M1/M2)

Settings

API Key:

Default Namespace: development

Test Connection



API EXPLORER

Request

GET /api/v1/status

Response

200 OK
{ status: 'active' }

Dogfooding:
These tools
are built as
Mini Apps.

DIAGNOSTICS (D1/D2)

Events Viewer

10:42:01 | navigate | {appId: 'settings'}

10:42:05 | api-request | GET /profiles

10:42:06 | html-loaded | {cacheKey: '...'}

Replay

Replay

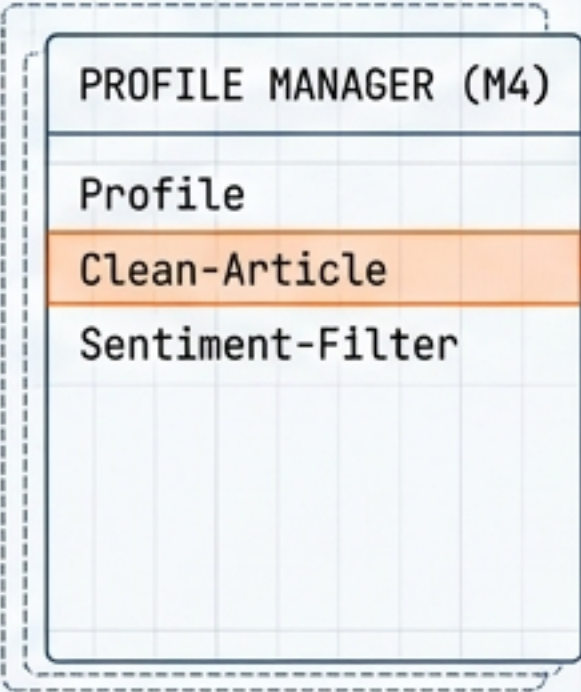
Replay

CORE WORKBENCH FEATURES

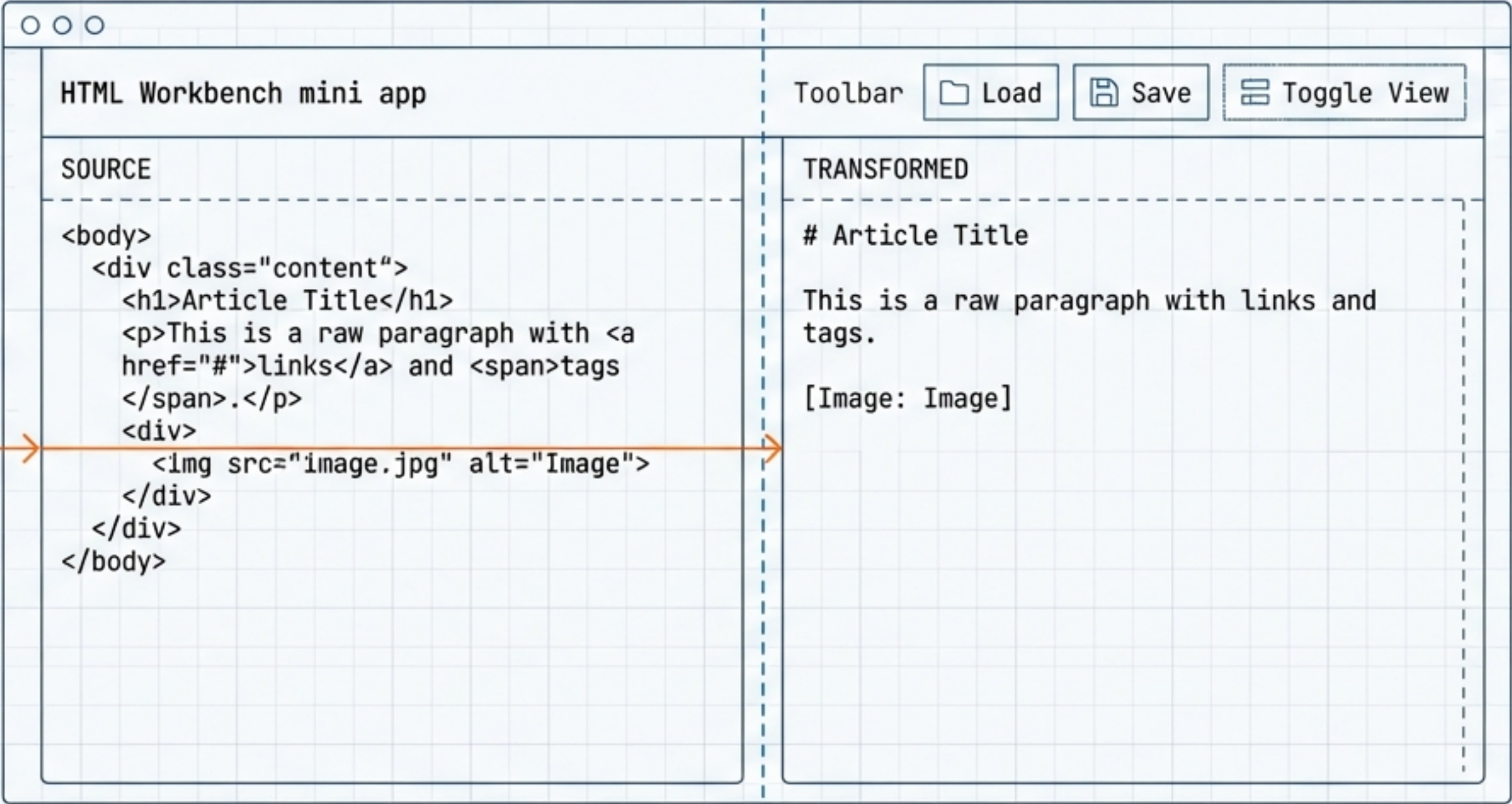
Stories M3, M4

HTML WORKBENCH (M3):
Split-pane visualisation.

Profile Manager (M4):
Feeds configuration.



Data Flow



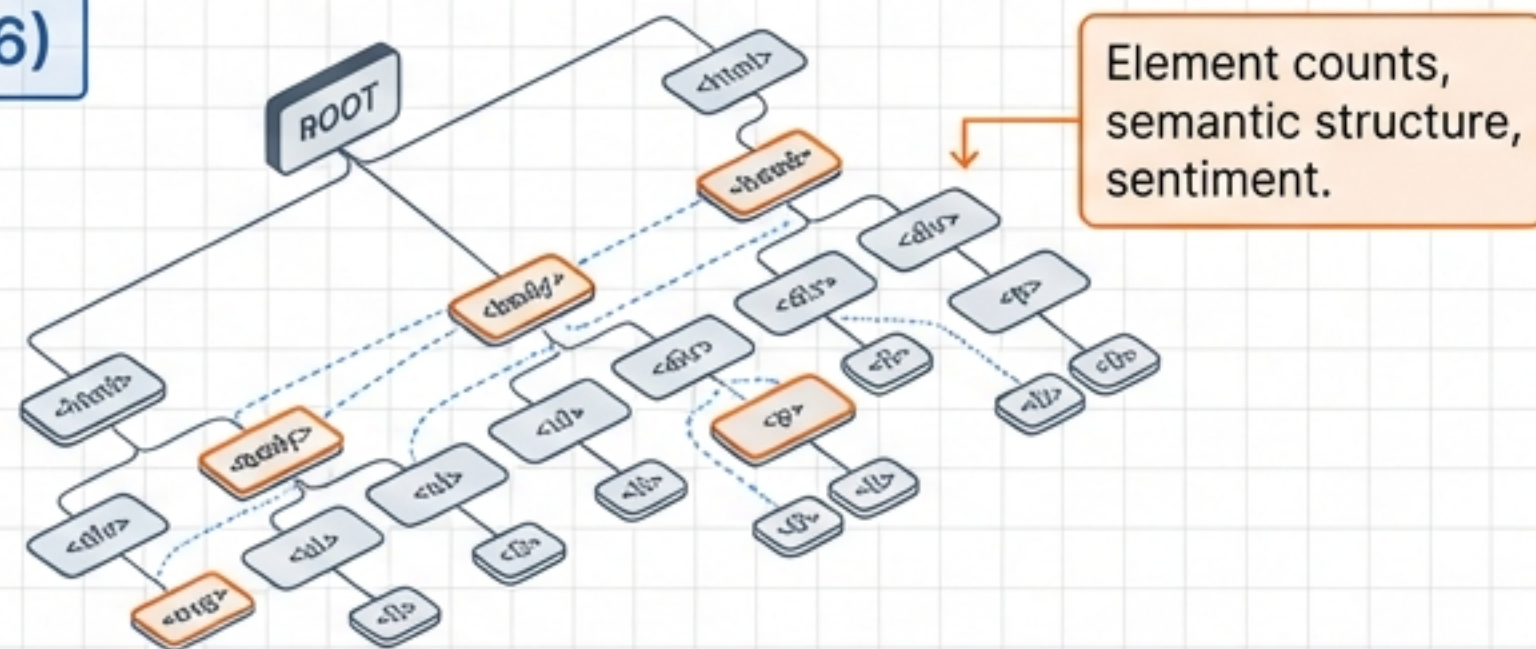
Shared Components: HTML Editor (S1), Transform Config (S3).

ANALYSIS & VISUALISATION

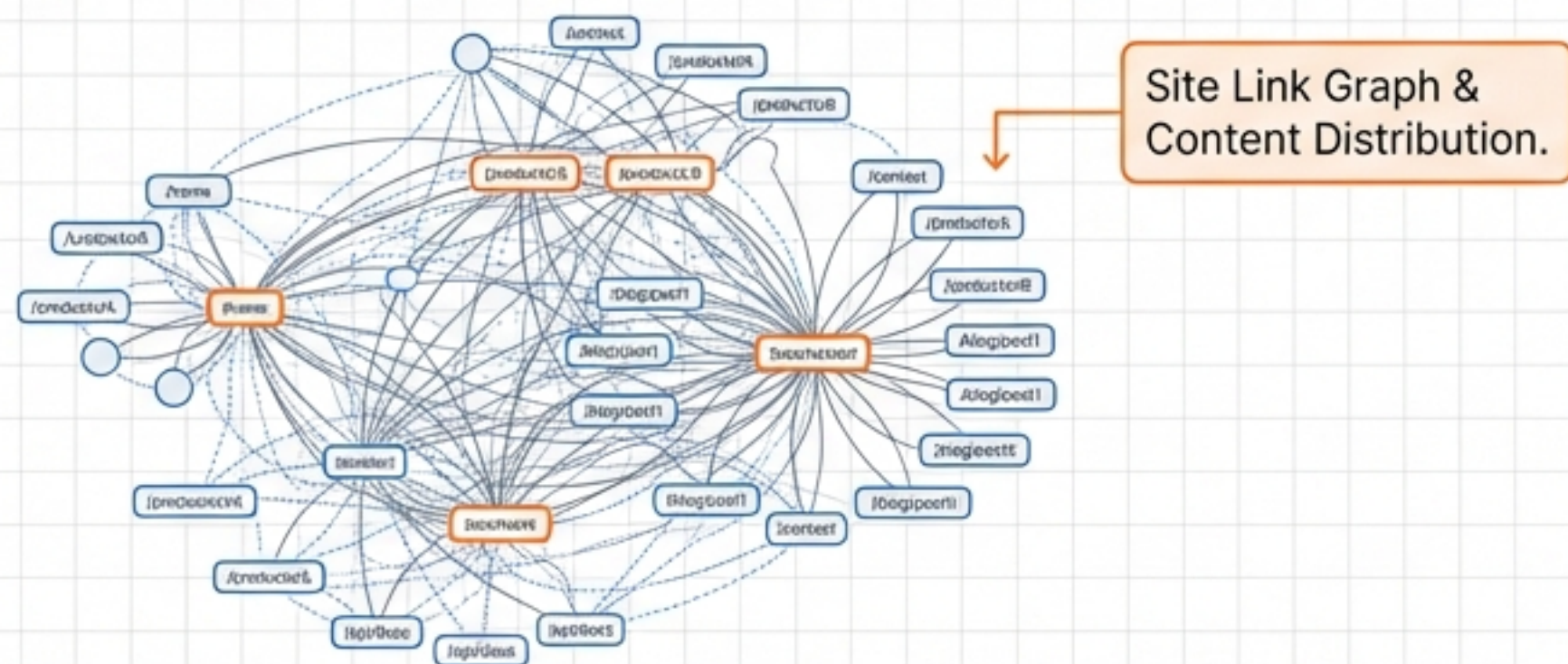
FEATURES

- **Site Browser (M5):**
Domain-level filtering.
- **Shared Component:**
Graph Visualiser (S2).

DOM Tree

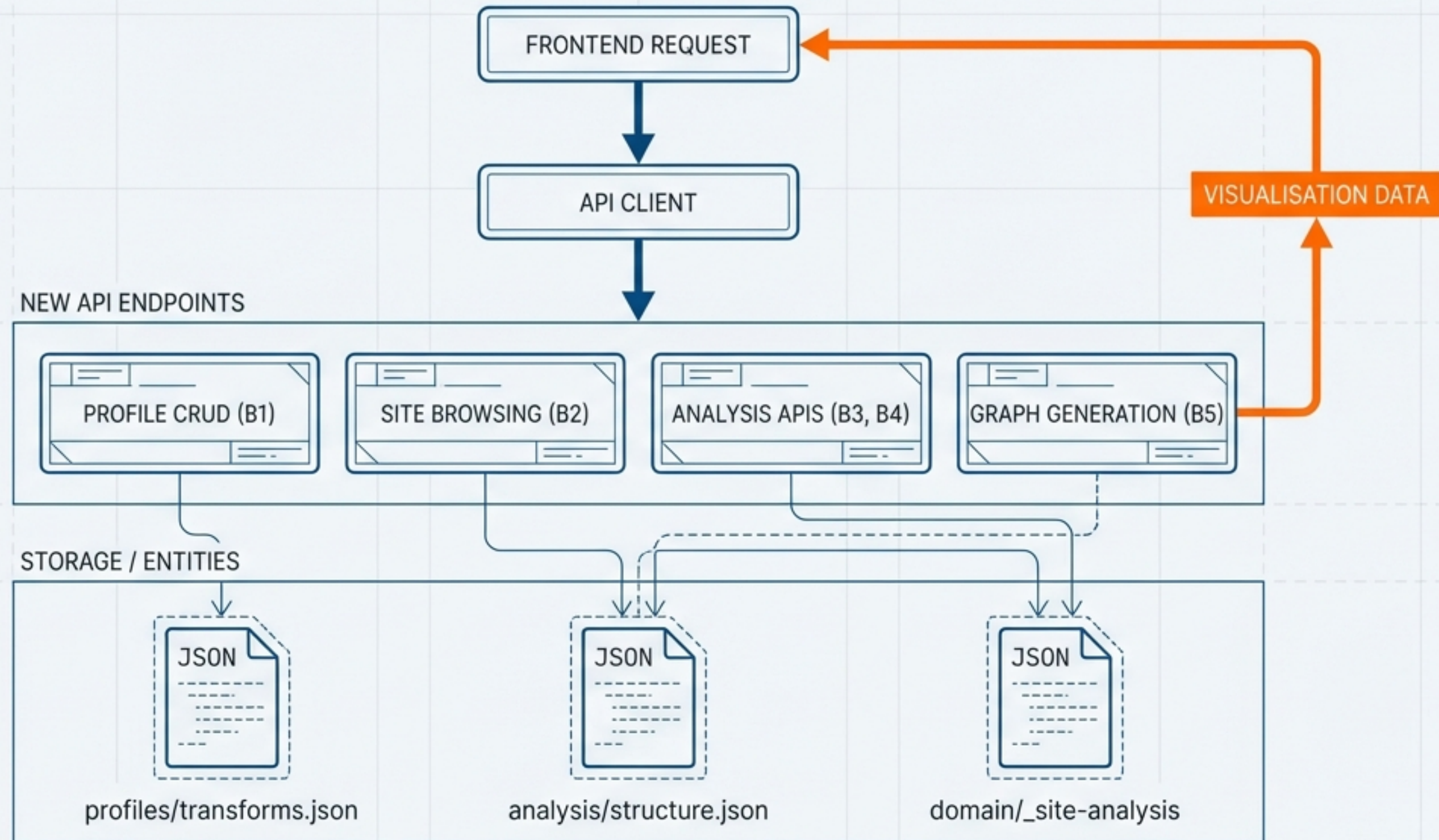


Force-Directed Graph

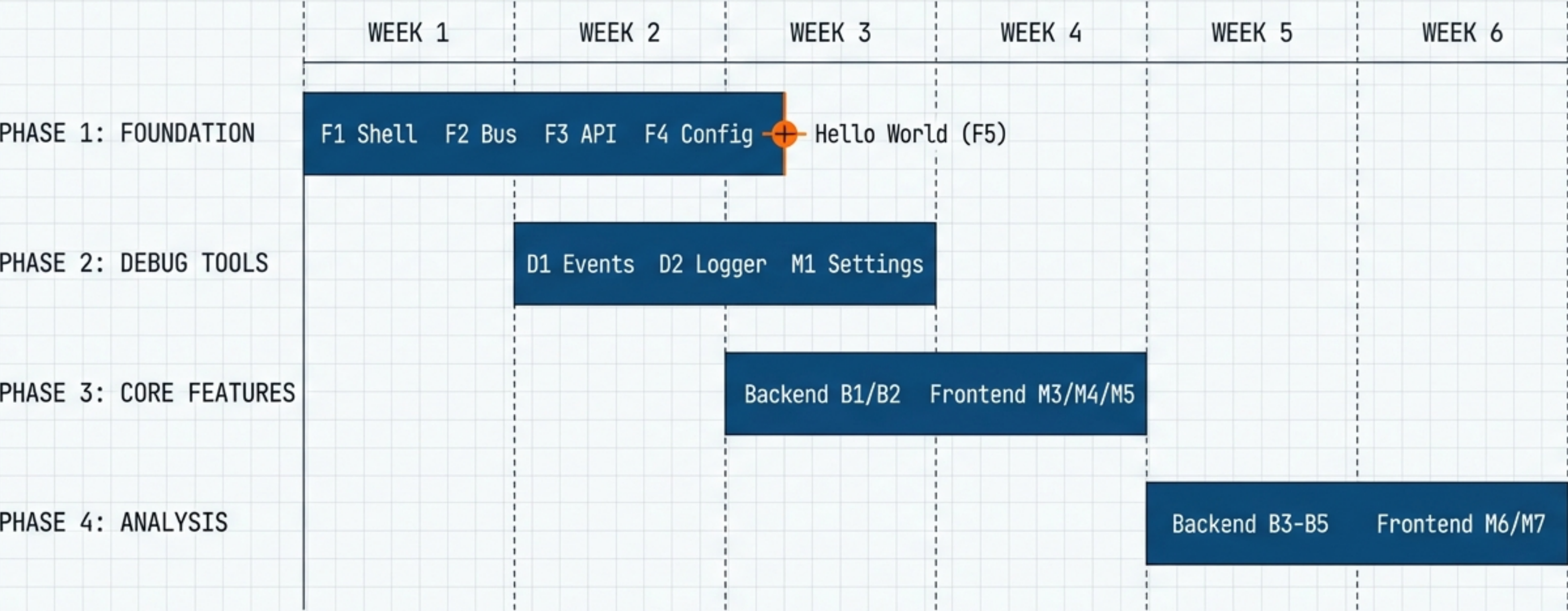


BACK-END INFRASTRUCTURE EXTENSIONS

Technical Slate: HTML Graph API Additions (Stories B1-B5)

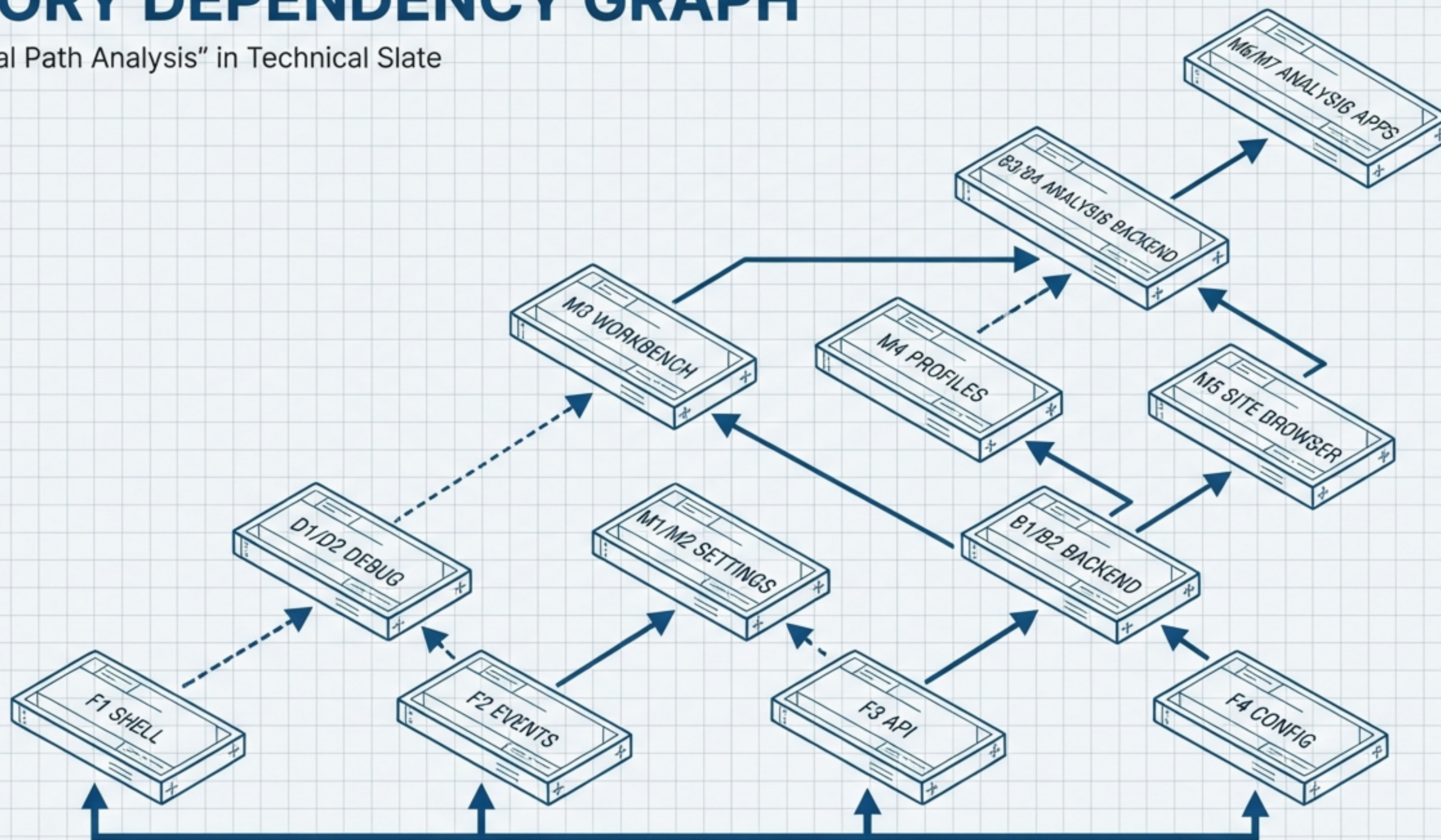


PHASED DEVELOPMENT STRATEGY



STORY DEPENDENCY GRAPH

"Critical Path Analysis" in Technical Slate



IMMEDIATE NEXT STEPS



A modular architecture designed for observability, scalability, and immediate value.